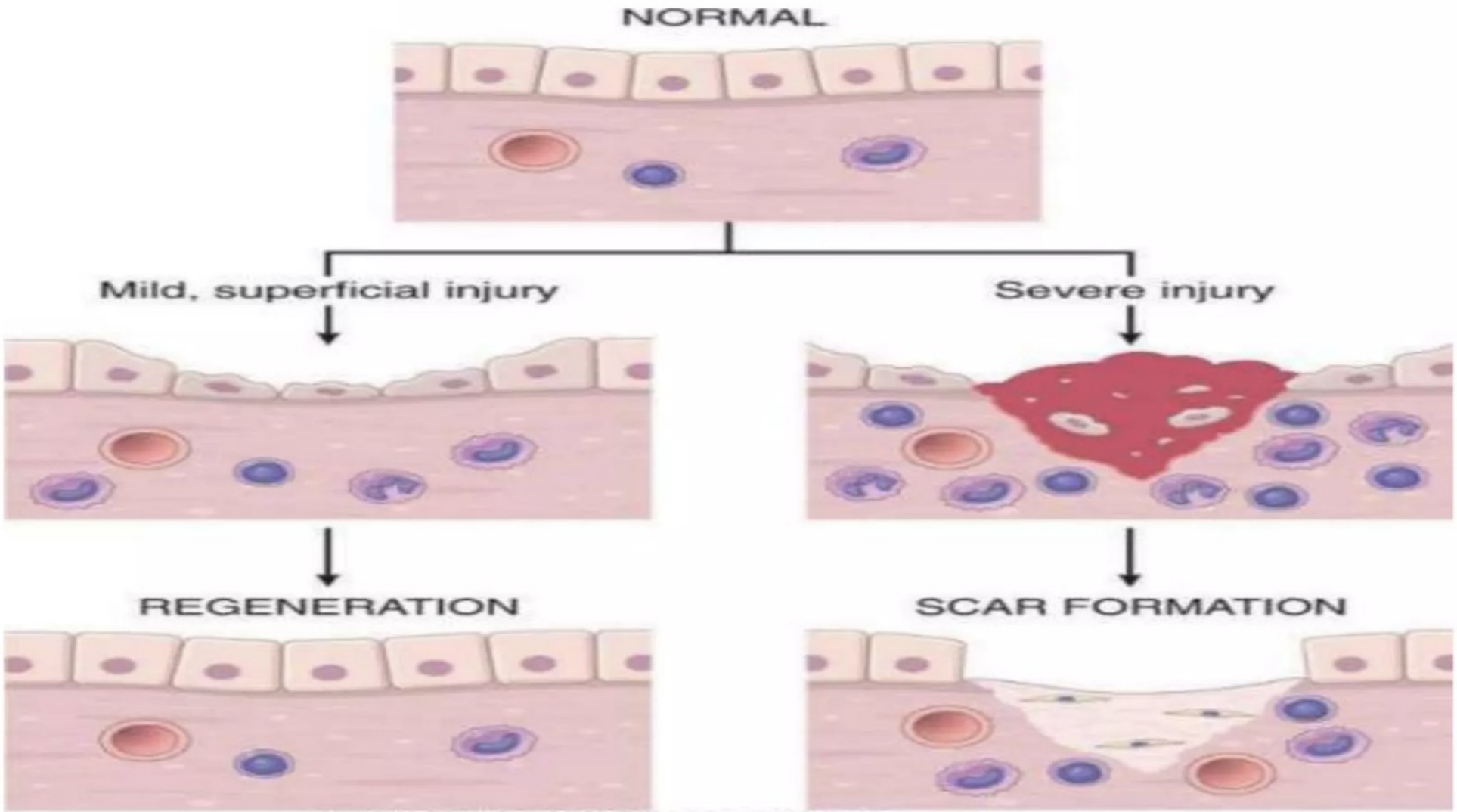
Healing process

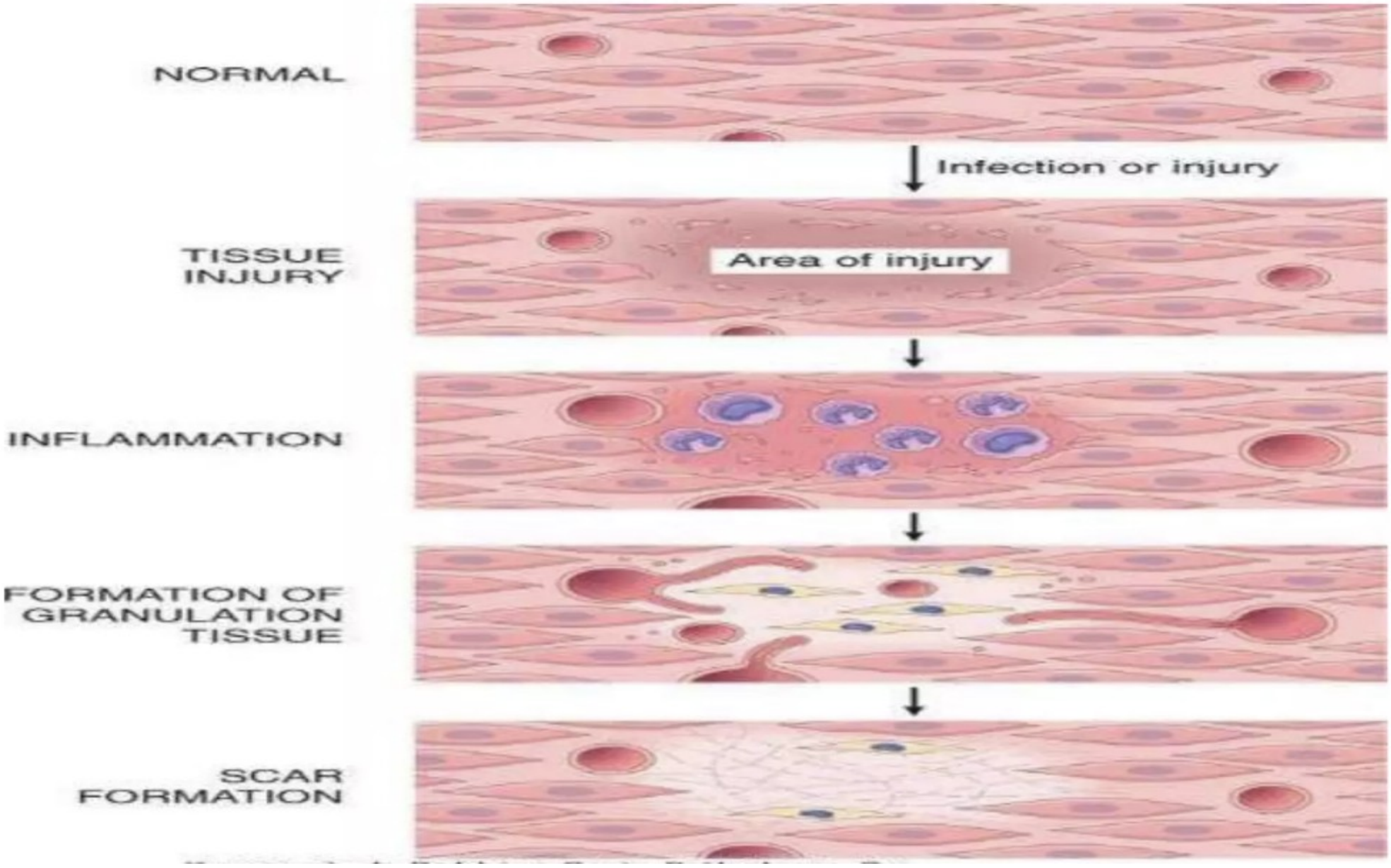
Learning outcome of healing process

- State the healing process.
- Type of cells involving in healing process
- Growth factor
- Regeneration and scar formation

Tissue Repair/Healing



Tissue Repair/Healing



Tissue Repair/Healing

Repair, sometimes called healing refers to the restoration of tissue architecture and function after an injury.

Repair of damaged tissues occurs by two types of reactions:

1. **Regeneration.**
2. **Connective tissue deposition (scar formation).**

Tissue Repair

- **Tissue repair = restoration of tissue architecture and function after an injury**
- **Occurs in two ways:**
  - **Regeneration of injured tissue**
  - **Replacement by connective tissue (scarring)**
- **Usually, tissue repair involves both processes.**
- **Involves cell proliferation, and interaction between cells and extracellular matrix.**

Regeneration

- Replacement of damaged components to return to **normal state (structure & function)** via cell proliferation of differentiated cells that still retain the capacity to proliferate, or via tissue stem cells / progenitors.
- Restoration of normal tissue structure can only occur if the **residual tissue is structurally intact** e.g. after partial surgical resection, in contrast to damage by infection or inflammation whereby scarring will also occur.
- i.e After removal of cellular debris, any necrotic parenchymal cells may be replaced by new parenchymal cells of the same type in a process known as regeneration.

- The ideal result of healing is to restore the tissue to its normal (preinjury) state, a process termed resolution
- Scar formation; if the injured tissues incapable of complete restitution, or if supporting structure severely damaged, repair occurs by laying down of connective (fibrous tissue)
- Organization; fibrosis +inflammatory exudate.

Organizing pneumonia

reparation

- restoration of normal shape x function is impaired or lost
- parenchyma replaced by fibrous tissue

The mechanism of healing depends on

- the type of inflammation,
- the extent of tissue necrosis,
- the types of cells involved, and
- the regenerative ability of damaged parenchymal cells.

- Lots of cells proliferate during tissue repair:
  - injured tissue remnants
  - vascular endothelial cells
  - fibroblasts
- We need to know about:
  - the cell cycle
  - the proliferative capacities of different tissues
  - stem cells
  - growth factors
  - the extracellular matrix

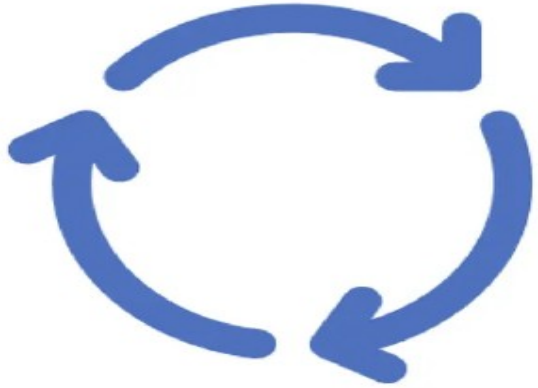
Cell cycle

Different proliferative capacity of the cells

Cell proliferation need Growth factors to promote the G0/G1 transitions & the entry of cells

Cell cycle is defined as the period between two successive cell divisions and is divided into 4 unequal chases:

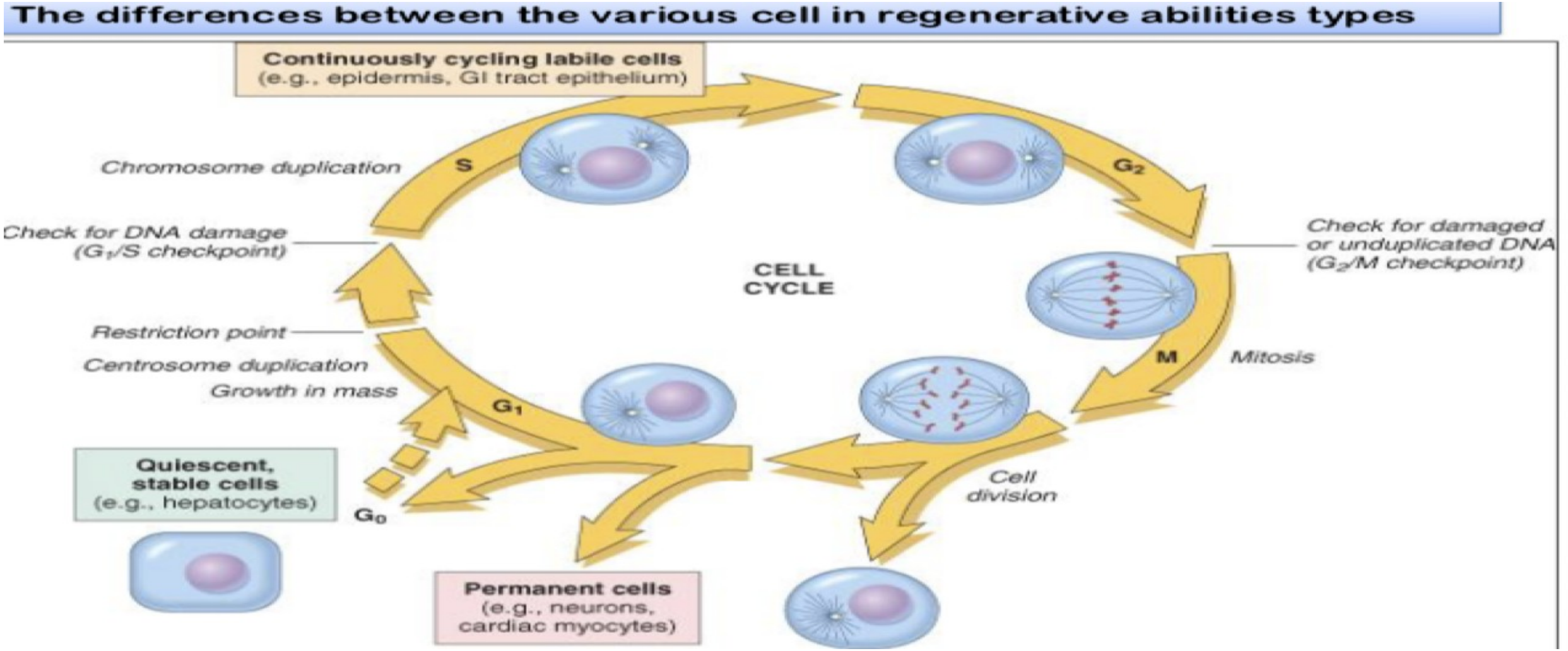
- M (mitosis) phase: phase of mitosis.
- G1, (gap i) phase: The daughter cell enters G1, phase after mitosis.
- S (synthesis) phase: During this phase, the synthesis of nuclear DNA takes place.
- G2, (gap 2) phase: After completion of nuclear DNA duplication, the cell enters G, phase.
- G0, (gap 0) phase: This is the quiescent or resting phase of the cell after an M phase.



Cell cycle

Different proliferative capacity of the cells

Cell proliferation need Growth factors to promote the G0/G1 transitions & the entry of cells



Tissue types

permanent
- nonproliferative in postnatal life
- neurons (?), cardiomyocytes (?)

stable
- regeneration as response to injury
- parenchyma – liver, pancreas, renal tubules
- mesenchymal cells, endothelium

Cells of these tissues are quiescent (in the G0 stage of the cell cycle) with minimal proliferative activity in the normal state, but can divide in response to injury or loss of tissue mass

labile
- continuous regeneration from stem cells (self-renewal)
- hematopoietic cells in bone marrow
- surface epithelia – skin, oral cavity, vagina, cervix
- duct epithelia – salivary glands, pancreas, biliary tract
- mucosas – GIT, uterus, fallopian tubes, urinary bladder

Growth factor

- Healing involves an orderly sequence of events which includes regeneration and migration of specialized cells, angiogenesis, proliferation of fibroblasts and related cells, matrix protein synthesis and finally cessation of these processes.
- These processes are mediated by **growth factors**.

Growth Factors

- Very important in tissue repair.
- Actions:
  - stimulate cell division and proliferation
  - promote cell survival
- Huge list! Usually have “GF” in name:
  - EGF
  - TGF
  - PDGF

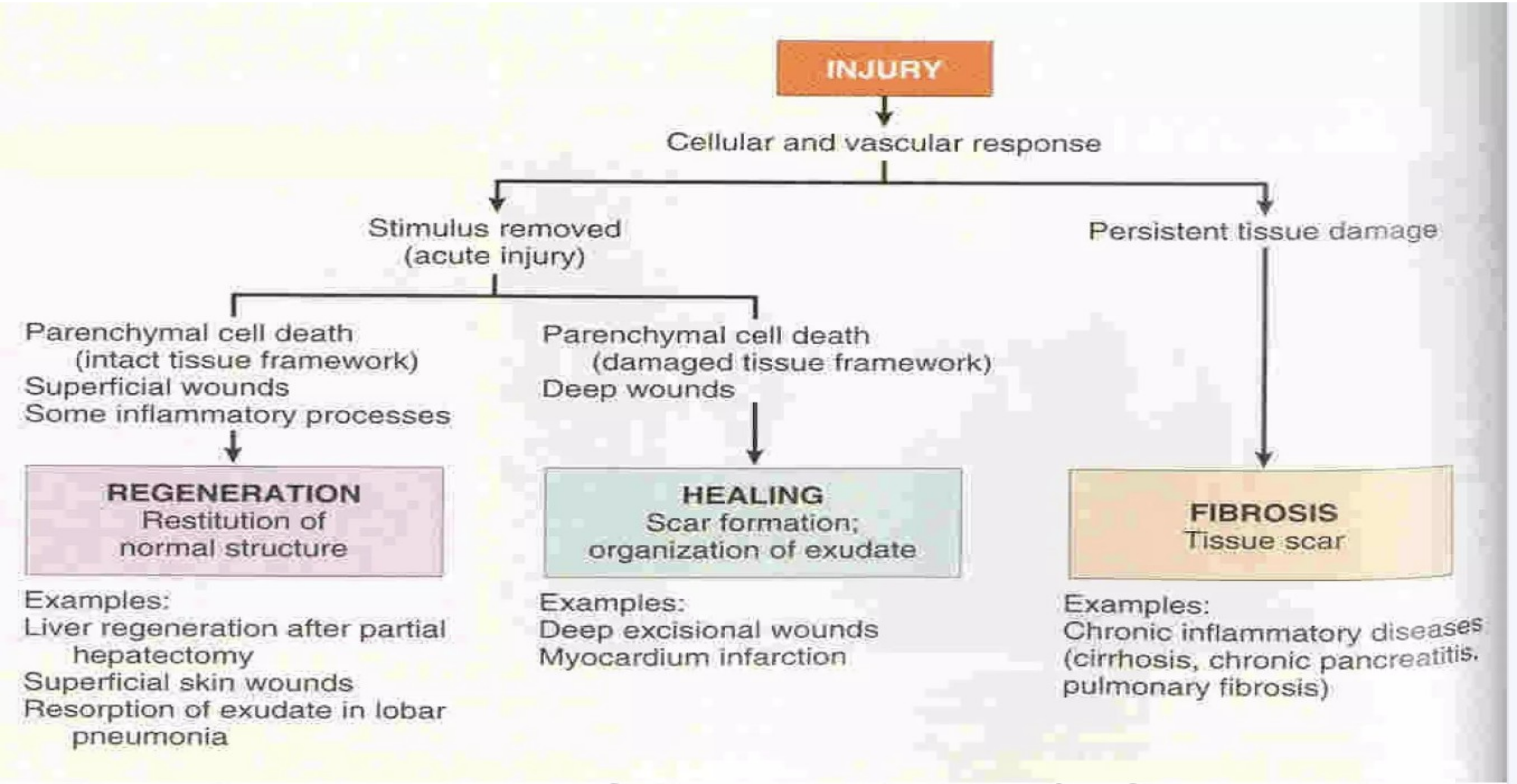
Sources of Growth Factors:

- Following injury, growth factors may be derived from a number of sources such as:

1. Platelets, activated after endothelial damage,
2. Damaged epithelial cells,
3. Circulating serum growth factors,
4. Macrophages, or
5. Lymphocytes recruited to the area of injury

Growth factors

- VEGF----Angiogenesis (migration & proliferation of endothelial cells)
- TGF-B (supress endothelial proliferation). Stabilize newly formed blood vessels
- PDGF (Recruitment of smooth muscle cell)
- ECM ---- (Angiogenesis)
- MMPs---- (ECM enzymes) permit remodeling & extension of vascular tube



Repair responses following Injury/Inflammation

Components of wound healing

1. Inflammation.

⇩

2. Proliferation and migration of parenchymal and connective tissue cells.

⇩

3. Formation of granulation tissue.

⇩

4. Wound contraction.

⇩

5. Acquisition of wound strength.

Following tissue injury, sequential phases for healing

- 1- Starting with a **hemostatic plug** composed of platelets to stop the bleeding and provide a scaffold for fibrin deposition.
- 2- Then after **inflammation** occurs and resolves,
- 3- various cell types (epithelial cells, endothelial cells and pericytes, fibroblasts) **proliferate** and migrate to close the now clean wound, initially via formation of **granulation tissue** (loose connective tissue with new thin-walled delicate capillaries and often admixed inflammatory cells) and
- 4- subsequent replacement by increasing amounts of collagen to form a **fibrous scar (maturation phase)**

Replacement of necrotic tissue

resorption by macrophages

dissolution by enzymes

replacement by granulation tissue

- uniform mechanism irrespective of inicial trigger
- the same microscopic appearance

- angiogenesis
- migration and proliferation of fibroblasts
- deposition of ECM
- maturation and reorganization

Repair by Connective Tissue Deposition

- **If repair cannot be accomplished by regeneration alone**
  - **it occurs by replacement of the injured cells with connective tissue,**
    - **leading to the formation of a scar,**
    - **or by a combination of regeneration of some residual cells and scar formation.**

❖ Scarring;

Deposition of connective tissue (fibrosis), when the **tissues are incapable of regeneration or supporting structure is too damaged to support regeneration alone**.

❖ Purpose;

is to provide enough structural stability for the injured tissue to function, not restoration

❖ This process is also known as organization when the fibrosis occurs in a tissue space occupied by inflammatory exudate (e.g. organizing pneumonia in the lung)

Fibrosis in parenchymal organs

❖ responsible for organ dysfunction / failure

- End stage kidney dis.
- e.g. liver cirrhosis,
- fibrosing diseases of the lung

❖Mechanism of fibrosis (excessive deposition of collagen and other ECM components in tissue) is basically similar to scar formation in the skin during tissue repair.

Steps in Scar Formation

- Repair by connective tissue deposition consists of sequential processes that follow tissue injury and the inflammatory response.
- **Angiogenesis** is the formation of new blood vessels, which supply nutrients and oxygen needed to support the repair process.
- Newly formed vessels are leaky because of incomplete interendothelial junctions and because VEGF, the growth factor that drives angiogenesis, increases vascular permeability.
- This leakiness accounts in part for the edema that may persist in healing wounds long after the acute inflammatory response has resolved.
- **Formation of granulation tissue.** Migration and proliferation of fibroblasts and deposition of loose connective tissue, together with the vessels and interspersed leukocytes, form *granulation tissue*.
- The **term *granulation tissue*** derives from its pink, soft, granular gross appearance, such as that seen beneath the scab of a skin wound.
- Its **histologic appearance** is characterized by proliferation of fibroblasts and new thin-walled, delicate capillaries (angiogenesis), in a loose extracellular matrix, often with admixed inflammatory cells, mainly macrophages. Granulation tissue progressively invades the site of injury; the amount of granulation tissue that is formed depends on the size of the tissue deficit created by the wound and the intensity of inflammation.

- **Remodeling of connective tissue.** Maturation and reorganization of the connective tissue (remodeling) produce the stable fibrous *scar*. The amount of connective tissue increases in the granulation tissue, eventually resulting in the formation of a scar, which may remodel over time.

Angiogenesis:

- Process of new blood vessel formation from existing vessels.
- Seen in healing, as well as development of collateral blood supply in ischemia and tumor growth.

Granulation tissue

pink soft granular appearance

richly vascularized

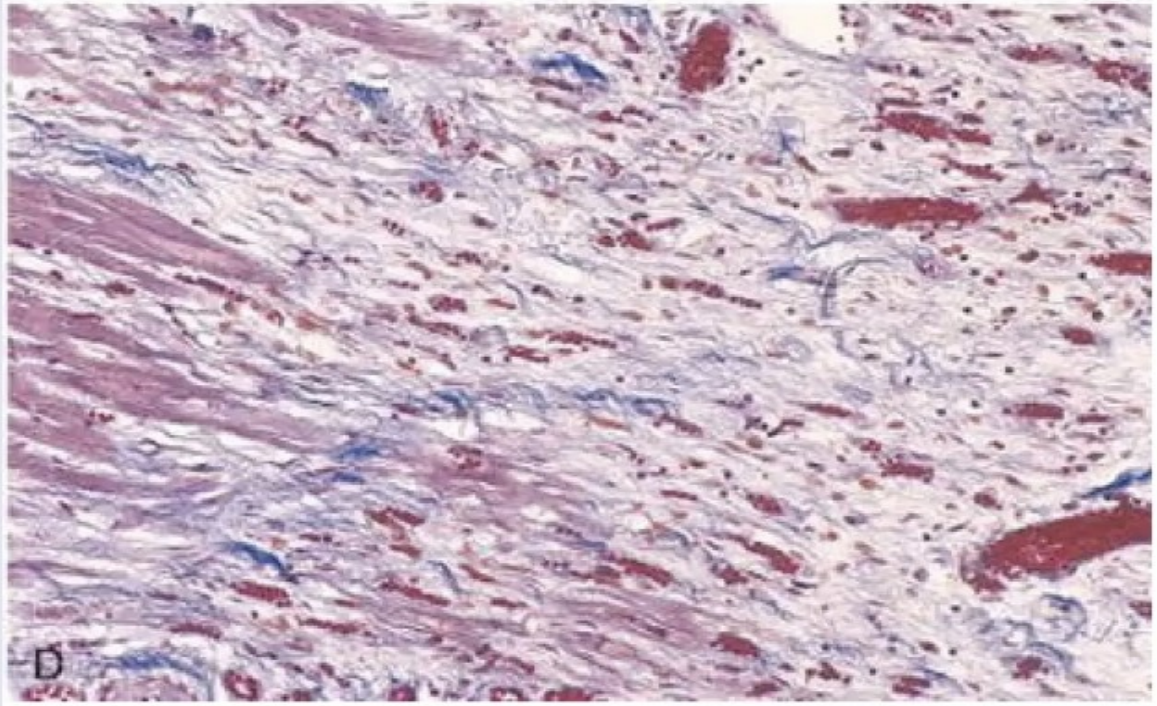
highly cellular

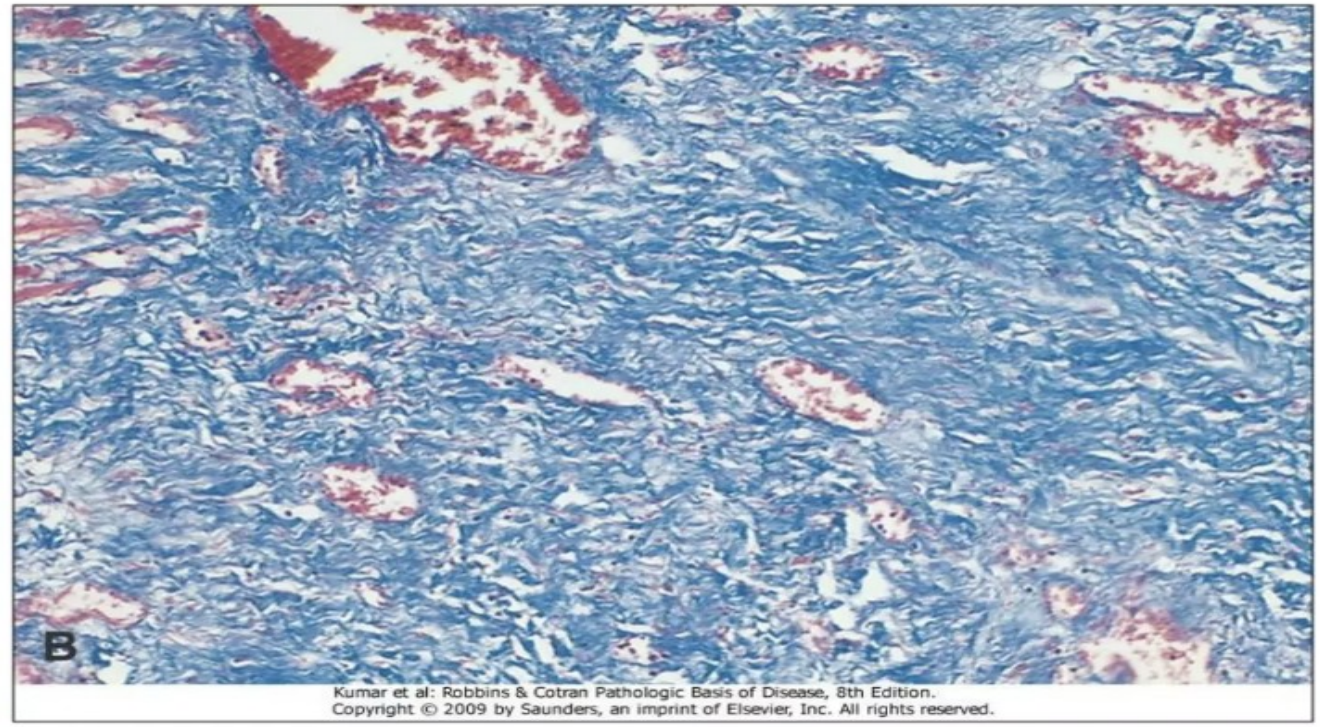
myxoid matrix

inflammatory cells

e.g. surface of wounds, bottom of ulcers







Kumar et al: Robbins & Cotran Pathologic Basis of Disease, 8th Edition.
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Granulation tissue showing numerous blood vessels, edema, and a loose extracellular matrix containing occasional inflammatory cells. Collagen is stained blue by the trichrome stain; minimal mature collagen can be seen at this point. **B,** Trichrome stain of mature scar, showing dense collagen, with only scattered vascular channels.

Thank You